

A QUICK NOTE ON INTEGRATION BY PARTS

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Recall the integration by parts formula:

$$\int u \, dv = uv - \int v \, du.$$

Neat, but the use of this formula requires mapping u and v to the specific integrand. This can be cumbersome, especially when the integrand is a product of three or more functions. Now, consider a seemingly more complicated but equivalent formulation:

$$\int xy \, dz = x \int y \, dz - \int \left(\int y \, dz \right) dx,$$

which turns out to be easier to apply in practice. Why? Because it reduces the amount of mental book-keeping: the right-hand side can be written directly from the left-hand side. Specifically, given the pattern

$$\int x (\text{something}) \, dz,$$

we can rewrite it immediately as

$$x \int (\text{something}) \, dz - \int \left(\int (\text{something}) \, dz \right) dx.$$

Happy integrating by parts!

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